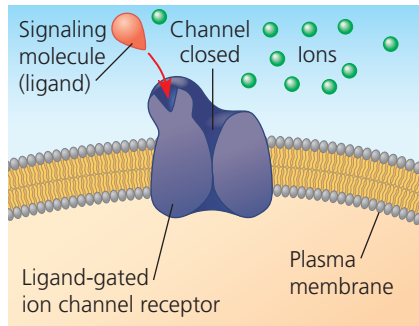


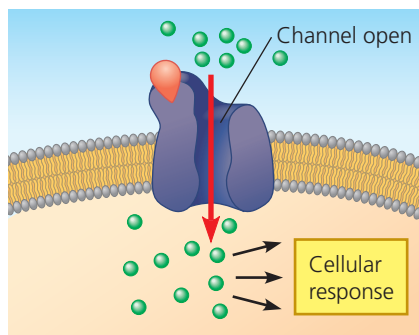
Ion Channel Receptors

A **ligand-gated ion channel** is a type of membrane channel receptor containing a region that can act as a “gate,” opening or closing the channel when the receptor changes shape. When a signaling molecule binds as a ligand to the channel receptor, the channel opens or closes, allowing or blocking the flow of specific ions, such as Na^+ or Ca^{2+} . Like the other receptors we have discussed, these proteins bind the ligand at a specific site on their extracellular sides.

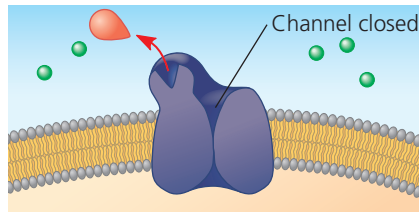
- 1 Here we show a ligand-gated ion channel receptor in which the channel remains closed until a ligand binds to the receptor.



- 2 When the ligand binds to the receptor and the channel opens, specific ions can flow through the channel and rapidly change the local concentration of that ion inside the cell. This change may directly affect the activity of the cell in some way.



- 3 When the ligand dissociates from this receptor, the channel closes and ions no longer enter the cell.



Ligand-gated ion channels are very important in the nervous system. For example, the neurotransmitter molecules released at a synapse between two nerve cells (see Figure 11.5b) bind as ligands to some ion channels on the receiving cell, causing the channels to open. Ions flow in (or, in some cases, out), triggering an electrical signal that propagates down the length of the receiving cell. Some of the gated ion channels are controlled by electrical signals instead of ligands; these voltage-gated ion channels are crucial to the functioning of the nervous system, as you'll see in Chapter 48. Some ion channels are present on membranes of organelles, such as the ER.

MAKE CONNECTIONS Is the flow of ions through a ligand-gated channel an example of active or passive transport? (Review Concepts 7.3 and 7.4.)

➔ **Mastering Biology Animation: Acetylcholine Receptor**

shown in Figure 11.7. In that case, the β -adrenergic receptor was stable enough to be crystallized only while it was among membrane molecules and in the presence of a molecule mimicking its ligand. Newer techniques that do not require crystallization, like cryo-EM (see Figure 6.3), have shown promise in determining the structures of cell-surface receptors.

Abnormal functioning of receptor tyrosine kinases (RTKs) is associated with many types of cancers. For example, some breast cancer patients have tumor cells with excessive levels of a receptor tyrosine kinase called HER2 (see Concept 12.3 and Figure 18.27). Using molecular biological techniques, researchers have developed a protein called Herceptin that binds to HER2 on cells and inhibits cell division, thus thwarting further tumor development. In some clinical studies, treatment with Herceptin improved patient survival rates by more than one-third. One goal of ongoing research into these cell-surface receptors and other cell-signaling proteins is development of additional successful treatments.

➔ **Mastering Biology Interview with Diana Bautista: How cell signaling leads to itch and pain (see the interview before Chapter 6)**



Intracellular Receptors

Intracellular receptor proteins are found in either the cytoplasm or nucleus of target cells. To reach such a receptor, a signaling molecule passes through the target cell's plasma membrane. A number of important signaling molecules can do this because they are either hydrophobic enough or small enough to cross the hydrophobic interior of the membrane (see Concept 7.1). Hydrophobic signaling molecules include steroid hormones and thyroid hormones of animals. Another chemical signaling molecule that possesses an intracellular receptor is nitric oxide (NO), a gas; this very small molecule readily passes between the membrane phospholipids. Once a hormone or other signaling molecule has entered a cell, its binding to an intracellular receptor changes the receptor into a hormone-receptor complex that is able to cause a response—in many cases, the turning on or off of particular genes.

The behavior of aldosterone is a representative example of how steroid hormones work. This hormone is secreted by cells of the adrenal gland (a gland that lies above the kidney), then travels through the blood and enters cells all over the body. However, a response occurs only in kidney cells because they alone contain receptors for aldosterone. In these cells, the hormone binds to and activates the receptor protein. With aldosterone attached, the active form of the receptor protein then enters the nucleus and turns on specific genes that control water and sodium flow in kidney cells, ultimately affecting blood volume (**Figure 11.9**).

How does the activated hormone-receptor complex turn on genes? Recall that the genes in a cell's DNA